



NATIONAL 5 ENGINEERING SCIENCE · MECHANISMS AND STRUCTURES

Pneumatics

Course notes

Version: 25 September 2026

Name _____

Class _____

Learning outcomes

Tick each box when you feel confident you can do it. Come back to this page before your unit test.

BY THE END OF THIS TOPIC...

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- | | | |
|----|---|--------------------------|
| 1 | I can describe the advantages and disadvantages of pneumatic systems, and the main safety rules. | ☐ |
| 2 | I can recognise and draw the symbols for pneumatic components. | ☐ |
| 3 | I can describe how a 3/2 valve works, including the functions of ports 1, 2 and 3. | ☐ |
| 4 | I can describe how a single-acting cylinder (SAC) is controlled by a 3/2 valve. | ☐ |
| 5 | I can design circuits using a T-piece, AND control (valves in series) and OR control (a shuttle valve). | ☐ |
| 6 | I can explain how a uni-directional restrictor controls the speed of a cylinder. | ☐ |
| 7 | I can describe how an air bleed circuit works. | ☐ |
| 8 | I can describe how a double-acting cylinder (DAC) is controlled by a 5/2 valve. | ☐ |
| 9 | I can describe the operation of semi-automatic and fully automatic circuits. | ☐ |
| 10 | I can explain how a reservoir and restrictor create a time delay. | ☐ |
| 11 | I can describe the difference between pilot-operated and solenoid-operated valves. | ☐ |
| 12 | I can calculate pressure, force and area, including the instroke force of a double-acting cylinder. | ☐ |

Where to write your answers: if you have access to a device, use the online course notes — you can type your answers into the answer panels and they are saved as you go. Otherwise, write your answers in your jotter, with the task and question number beside each answer.

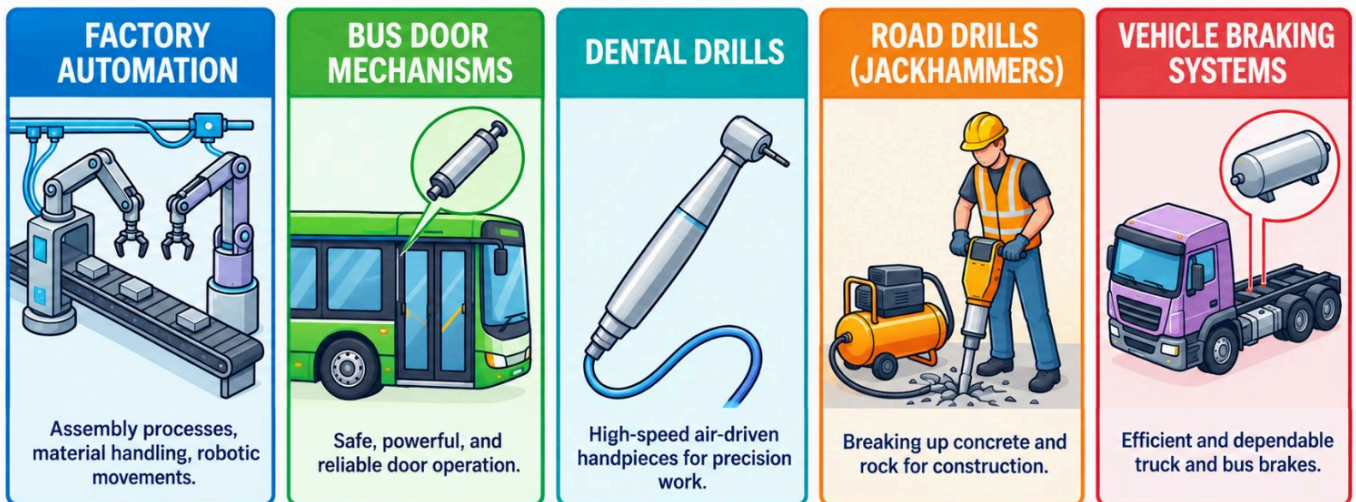


Scan for the online
course notes

Section 1: What are Pneumatics?

Pneumatics is the use of **pressurised air** to do mechanical work. A **compressor** draws in air from the atmosphere and compresses it to a higher pressure. This compressed air is stored and then controlled to produce linear or rotary movement. Pneumatic systems are found all around us — in factory automation, bus door mechanisms, dental drills, road drills, and vehicle braking systems.

PNEUMATIC SYSTEMS ARE FOUND ALL AROUND US POWERING MANY COMMON MACHINES AND DEVICES WITH COMPRESSED AIR



Key point: Pneumatic systems are widely used in industry because compressed air is safe, clean, and readily available.

Safety: Isolate the air supply before connecting or disconnecting components. Never point a pneumatic line at anyone. Check all connections are secure before pressurising a circuit.

Advantages of pneumatic systems

Pneumatic systems offer several advantages over hydraulic or electrical systems: compressed air is **non-flammable** and safe to use in hazardous environments; the components are relatively low-cost; speed is easy to adjust using flow control valves; and any leaks result only in air escaping rather than fluid spillage.

Disadvantages

Pneumatic systems can be noisy, and energy is lost as heat when air is compressed. A compressor and air supply are always required. Pneumatic systems are usually limited to two fixed positions, set by mechanical stops — accurately stopping a piston partway between these stops is difficult, since air's compressibility lets it drift rather than staying rigidly in place.

Task 1 — Pneumatics in context

1a. Give three examples of pneumatic systems you might encounter in everyday life. For each one, describe what work the compressed air is doing.

1b. Explain two advantages and two disadvantages of using a pneumatic system compared to an electric motor for a door-opening mechanism.

Section 2: Pneumatic Components — Seeing it Work First

Rather than starting with a large number of symbols to memorise, let's watch a real circuit working first. Seeing the simulation below will give you a feel for what's going on — the air flowing, the valve switching, the cylinder moving — before we break down exactly *why* it behaves that way and what each part is called.



No Pressure Sim — Single Acting Cylinder (SAC)

This interactive simulator is in the online course notes. Scan the QR code, or use a device, to open it.

Clicking the push-button valve lets main air flow through to the cylinder, pushing the piston out — the **outstroke**. Releasing the button cuts off that air supply, and a spring inside the cylinder pushes the piston back in — the **instroke** — venting the used air out the other side of the valve as it goes.

Now let's look at what you just saw, piece by piece.

Main air supply and exhaust

Every pneumatic circuit needs a **main air supply** — the pressurised air from the compressor, connected into port 1 of a valve. **Exhaust** is where used air vents to atmosphere after passing through the circuit, shown in diagrams as an open triangle.



Compressor
Produces the main air supply



connection to
component

Main Air Supply
Enters at port 1



connection from
component

Exhaust Port
Vents used air to atmosphere

The 3/2 valve

The valve you just operated is a **push-button, spring-return 3/2 valve** — it has 3 ports and 2 states.

We have to be specific about the full name, because a 3/2 valve can be switched from its normal state to its actuated state in several different ways. The number of ports and states tells you *what* the valve does; the actuator tells you *how* it gets triggered. The same 3/2 valve could instead be built as a roller 3/2 valve, a lever 3/2 valve, a pilot 3/2 valve, and so on — the switching behaviour is identical, but what causes the switch is different.

The word "actuate": in pneumatics, to **actuate** a valve means to trigger it to move from its normal (rest) state into its other state. The **actuator** is whatever causes that triggering to happen — a finger on a button, a cam tripping a roller, a small air pressure signal, and so on. A valve is described by both its ports/states *and* its actuator, e.g. "push-button, spring-return 3/2 valve".

Here are the actuators you'll meet across these notes:

 Push-button Pressed by hand	 Spring Return Returns the valve to normal	 Roller Actuated while a moving part presses on it
 Roller Trip Actuated momentarily as a part passes over it	 Lever Operated by hand (or foot) and usually latches on	 Plunger Pushed end-on
 Pilot Switched by main air pressure	 Diaphragm Switched by a small pressure change See Section 7	 Solenoid Switched electrically See Section 12

Here's what's actually happening inside a push-button, spring-return 3/2 valve as you press and release the button:

WORKED EXAMPLE — HOW THE 3/2 VALVE SWITCHES

- 1 **At rest:** The valve is in its normal (spring-return) state. Port 1 (main air) is blocked; port 2 (cylinder) is connected to port 3 (exhaust). The cylinder spring holds the rod instroke.
- 2 **Button pressed:** The valve shifts to its actuated state. Main air at port 1 now connects to port 2 (cylinder). Port 3 (exhaust) is blocked. Pressurised air causes the outstroke against the spring.
- 3 **Button released:** The valve spring returns it to its normal state. Port 2 connects back to port 3 (exhaust). Air vents to atmosphere and the cylinder spring instrokes the rod.

Single acting cylinder (SAC)

The cylinder in the simulation is a **single acting cylinder (SAC)**. Main air causes the outstroke (extension of the piston rod); a spring returns it when the pressure is released. SACs are used where force is only needed in one direction.



Schematic drawing: You've just met your first valve and cylinder. In your assessment, you will not need to draw a component completely from scratch — but you will be given circuit layout diagrams to read, and you'll often need to **complete a partly-drawn circuit**, for example adding in missing piping between components.

It is therefore useful to be able to recognise symbols and know how they are connected — and the best way to learn to do this is to practice drawing the symbols and circuits yourself. That's why these notes include drawing tasks marked  throughout. For these, **draw your answer by hand in your jotter**.

Task 2 — Main air, exhaust, the 3/2 valve and the SAC

2a. Use the No Pressure simulator or physical components to build the SAC circuit. Operate the push-button valve and describe what happens. In your answer, describe the state of the valve and what the cylinder does when the button is pressed and when it is released.

2b. Describe what happens to the ports of a 3/2 valve when it switches from its normal state to its actuated state.

2c.  Draw the full circuit diagram for the SAC circuit: a 3/2 push-button valve (normal position) connected to a single acting cylinder, with main air and exhaust labelled. Zoom into the symbols in the simulator above to check the details of how to draw them. Draw your answer in your jotter.

Section 3: T-Piece — Splitting the Air Supply

A **T-piece connector** lets one main air supply feed two or more branches of a circuit at once. See it in action below.



No Pressure Sim — T-Piece Air Supply Splitting

This interactive simulator is in the online course notes. Scan the QR code, or use a device, to open it.

Both branches receive the same supply pressure. T-pieces are used when a single air source needs to power multiple cylinders or valves at the same time.

Task 3 — T-piece

3a. Use the simulator or physical components to build a circuit using a T-piece to operate two cylinders from the same main air supply. Describe what you observe. What happens to each cylinder when the valve is operated?

3b. Give one example of a practical application where a T-piece would be useful in a pneumatic circuit.

Section 4: Shuttle Valve (OR control)

A **shuttle valve** lets a cylinder be operated from two separate control valves — actuating either one, or both together, causes the outstroke. This is known as **OR control**.



No Pressure Sim — Shuttle Valve

This interactive simulator is in the online course notes. Scan the QR code, or use a device, to open it.

Inside the shuttle valve is a small ball. When main air enters from one inlet, it pushes the ball across to seal the other inlet and passes through to the outlet. If main air enters the other inlet instead, the ball moves the other way and the same thing happens. If main air enters both inlets at the same time, the ball stays where it is and main air still passes through to the outlet.

You might wonder why a plain T-piece couldn't be used here instead, since it also joins two pipes together. A T-piece has no ball or seal inside it — it's just an open junction. If only *one* of the two valves were actuated, the main air would take the path of least resistance and simply escape out through the **other valve's exhaust port**, instead of reaching the cylinder. A shuttle valve prevents this by sealing off whichever inlet isn't currently supplying air.

Key point: A shuttle valve lets the cylinder outstroke if valve A or valve B (or both together) supplies main air — this is called **OR control**. It's useful for two-handed operation or operating a machine from two different locations.

Task 4 — Shuttle valve

4a. Explain how a shuttle valve works. What happens inside the valve when main air enters from one inlet? What happens if main air enters from both inlets at the same time?

4b. A press in a workshop can be dangerous to operate with one hand. Describe how a shuttle valve could be used as part of a two-handed safety circuit.

4c.  Draw the circuit symbol for a shuttle valve, clearly showing its two inlets and single outlet. Draw your answer in your jotter.

Section 5: AND Control

Section 4 covered **OR control** — where valve A or valve B (or both valves) are actuated to outstroke a cylinder. Sometimes you want a cylinder to only outstroke if **two conditions are both satisfied at once**. This is called **AND control**.

Where AND control is used

The classic example is a **two-handed safety control** on a machine like a guillotine or a press. If the machine only needed one button pressed, an operator could reach into the danger area with their free hand while pressing the button with the other. Setting up the machine so that **both** buttons — one for each hand — must be pressed at the same time guarantees both of the operator's hands are safely out of the way before the machine can move.

Other examples include a machine guard that must be closed **and** a start button pressed before a cylinder will move, or a security gate that needs a card swipe **and** a button press together.

Key point: OR control needs a specific component (a shuttle valve) but AND control can be built using two ordinary 3/2 valves, but they have to be arranged in a specific way.

Task 5 — Design an AND control circuit

5a.  Using two 3/2 push-button, spring-return valves and a single acting cylinder, design a circuit where the cylinder will only outstroke when **both** push-buttons are pressed at the same time. Pressing only one button (either one) must **not** cause the outstroke. *Clue: you only need one main air supply.* Draw your circuit in your jotter.

5b. Build or simulate your design. Test what happens when you press only the first button, only the second button, and both together. Describe what you observed for each test.

5c. Explain, in terms of the path the main air has to take, why pressing only one of the two valves does not cause the cylinder to outstroke.

Section 6: Speed Control — Restrictors

In many applications it is necessary to control the speed that the cylinder moves. This is achieved by restricting the flow of air.

The restrictor

A **restrictor** slows a cylinder down by reducing the amount of space the air can flow through — a small adjustable screw narrows the passage, so less air can get through per second. The narrower you make it, the slower the cylinder moves.



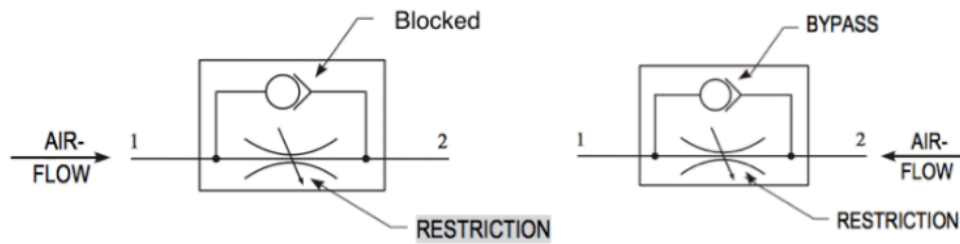
Restrictor

Adjustable flow control valve

Problem with a plain restrictor: it narrows the passage in *both* directions at once. That slows the cylinder down whichever way it's moving — outstroke *and* instroke — when we usually only want to control one direction. It can also make the piston's movement jerky rather than smooth.

The uni-directional restrictor (UDR)

A **uni-directional restrictor** solves this by combining the same adjustable restriction with a one-way **bypass channel** running alongside it. The bypass contains a small flap or ball that acts like a one-way valve:



- In the **left-hand diagram**, air is trying to flow from port 1 to port 2. The flap is pushed shut, blocking the bypass channel, so the air is forced through the narrow adjustable orifice — this direction is **restricted**.
- In the **right-hand diagram**, air is flowing the other way, from port 2 to port 1. This pushes the flap open, so the air flows straight through the wide bypass channel instead — this direction is **not restricted**.

That's why it's called *uni-directional*: free flow one way, restricted flow the other — set by the single adjustable screw.

Where to put it

A UDR is placed so that it restricts the air **leaving** the cylinder (the exhaust side), not the air coming in. If you restrict the incoming air instead, the piston has to fight to build up enough pressure to move at all, while the opposite side empties out freely — this makes the motion hesitant and jerky. Restricting the exhaust instead means main air can still push in freely, and the piston is held back smoothly by air trapped behind it, controlled by how fast it's allowed to escape.

Exam rule: In pneumatics, always restrict the exhaust air leaving a cylinder — never the main air going in. This is what gives smooth, controlled movement.



No Pressure Sim — SAC with Restrictors

This interactive simulator is in the online course notes. Scan the QR code, or use a device, to open it.

Task 6 — Speed control

6a. Use the simulator or physical components to build a circuit with a UDR. Adjust the restrictor and observe how the cylinder speed changes. Explain, using the idea of the bypass channel, why a UDR only restricts the airflow in one direction.

6b. A packaging machine uses a SAC to push items along a conveyor. The operator wants the outstroke to be slow but the instroke to be quick. Explain where a uni-directional restrictor (UDR) should be placed and how it should be oriented to achieve this.

6c.  Draw a SAC circuit with a uni-directional restrictor correctly positioned on the exhaust to slow the outstroke. Label the direction of free flow and restricted flow. Draw your answer in your jotter.

Section 7: Air Bleed (Diaphragm) Circuit

An **air bleed circuit** works differently from every circuit so far — instead of pressing a valve to supply air, it works by **blocking** a small continuous flow of air. Try it in the simulator before reading the detail.



No Pressure Sim — Air Bleed Circuit

This interactive simulator is in the online course notes. Scan the QR code, or use a device, to open it.

How it works

The circuit uses a **diaphragm-actuated** 3/2 valve. In the normal state, main air constantly bleeds to atmosphere through a small hole in the circuit (the bleed point). Because this air is always escaping, there is never enough pressure to act on the diaphragm — the valve stays in its normal state and the cylinder remains instroked.

A **uni-directional restrictor (UDR)** is fitted between the main air supply and the bleed nozzle. Adjusting it controls how freely air is allowed to escape at rest, which lets the circuit be **tuned to suit whatever is being sensed** — open it up for something that blocks the nozzle firmly and completely, or wind it in so that even the lightest, most fleeting obstruction is enough to build up pressure and switch the valve.

When an object **covers the bleed hole**, the escaping air is blocked. Pressure immediately builds up behind the diaphragm. Once it reaches the switching threshold, the diaphragm switches the 3/2 valve, main air flows to the cylinder, and the cylinder outstrokes.

As soon as the obstruction is removed, air bleeds away again, pressure drops, the diaphragm releases, and the valve spring returns it to its normal state — causing the cylinder to instroke.

Key point: The air bleed circuit is a **proximity/presence detection** circuit. No electrical signal is needed — physical contact or proximity blocking the bleed hole is enough to trigger the cylinder. This makes it useful for detecting products on a conveyor, detecting when a hand is near a machine, or triggering an action when a part arrives at a station. Because it needs only a change in air pressure and no physical contact from a sensor, it's especially useful for detecting **delicate or fragile items** — such as eggs or electronic circuit boards — where a mechanical touch sensor could easily damage them.

Task 7 — Air bleed circuit

7a. Explain how an air bleed circuit works. In your answer, describe what happens at the bleed point when it is open, and what happens when it is blocked.

7b. Describe one practical application of an air bleed circuit. Explain what causes the bleed to be blocked in your example and what action this triggers.

7c. State one key difference between an air bleed circuit and a standard push-button SAC circuit.

7d.  Draw the full air bleed circuit: including the diaphragm-actuated 3/2 valve, the bleed point, main air, exhaust, and the single acting cylinder. Try to do this as best you can from memory before you look at the simulator or solution below. Draw your drawing in your jotter — then check it against the diagram above.

Section 8: Double Acting Cylinder Circuit

So far, we've been working with cylinders that only need to be controlled in one direction. A spring has returned the piston to where it started. However, we often need to control both the outstroke and instroke of a cylinder. To do this, we use a double acting cylinder (DAC) that has a port at each end. Let's consider using 3/2 valves to directly control the air going into each end of a DAC.

The problem with two 3/2 valves

Imagine connecting one 3/2 push-button valve to the outstroke side of a DAC, and a second 3/2 push-button valve to the instroke side. Pressing the first button lets main air push the piston out; pressing the second lets main air push it back in.

This sounds reasonable, but it relies entirely on the operator pressing exactly one button at a time, in the right order. Press both together, or release both, and **neither** side of the piston is being actively driven — the cylinder just sits wherever it happens to be, with nothing holding it firmly in position. There's nothing in the circuit itself that prevents this.

What we actually need: a valve that guarantees exactly one side of the piston is connected to main air at all times, with the other side always exhausting — automatically, from a single action, without relying on an operator to coordinate two separate valves correctly.

That's exactly what a **5/2 directional control valve** does. Because it only ever has two possible states, and each state connects main air to one side while exhausting the other, it's physically impossible for both sides (or neither side) to be pressurised at once. Try it in the simulator below.



No Pressure Sim — Double Acting Cylinder (DAC)

This interactive simulator is in the online course notes. Scan the QR code, or use a device, to open it.

Key point: Air that has been controlled by a valve, and then goes on to control *another* valve, is known as **pilot air**. On a circuit diagram, pilot air is drawn as a **dashed line**, so you can tell it apart from the solid lines used for main air and exhaust.



A 5/2 valve has five ports and two states. In one state, port 1 (main air) connects to port 2 (outstroke side) while port 4 (instroke side) exhausts through port 5. Switching the valve reverses this: port 1 now connects to port 4, while port 2 exhausts through port 3. Changing the state of the valve therefore reverses the direction of the cylinder — and at every moment, one side is under pressure while the other is venting.

DAC advantage over SAC: Because main air actively drives the instroke as well as the outstroke, a DAC can produce a pushing force in both directions. The instroke force is slightly less than the outstroke force because the piston rod reduces the effective area on the instroke side (see Section 13).

Task 8 — DAC circuit

8a. Use the No Pressure simulator or physical components to build the DAC circuit. Operate the 5/2 valve in both directions. Describe what you observe and explain how the DAC differs from the SAC circuit you built in Task 2.

8b. A DAC circuit has its 5/2 valve in the initial state. Describe what the cylinder is doing. Explain what happens to the cylinder when the valve changes to its actuated state.

8c.  Draw the full circuit diagram for a DAC circuit: a 5/2 push-button valve operated by two 3/2 valves, connected to a double acting cylinder, with the ports numbered. Zoom into the symbols in the simulator above to check the details of how to draw them. Draw your answer in your jotter.

Section 9: Semi-Automatic Circuit

A **semi-automatic circuit** needs a manual input to start each cycle, but part of the operation happens automatically.



No Pressure Sim — Semi-Automatic Circuit

This interactive simulator is in the online course notes. Scan the QR code, or use a device, to open it.

The cylinder extends (outstroke), then automatically retracts (instroke) when it reaches the end stop. The circuit then waits for your next button press.

A typical arrangement uses a DAC, a 5/2 valve with **pilot actuators**, and a roller spring-return valve triggered by the cylinder rod at the end of its outstroke.

WORKED EXAMPLE — SEMI-AUTOMATIC CYCLE

- 1 Start:** Operator presses push-button → a pilot signal is sent to the 5/2 valve which causes it to change to its actuated state → main air flows to the outstroke side (port 2) of the DAC → cylinder outstroke begins.
- 2 End of outstroke:** Rod actuates the 3/2 roller spring-return valve → a pilot signal is sent to the 5/2 valve and it returns to its normal state → main air causes the DAC to instroke → outstroke side exhausts through port 3.
- 3 Wait:** Cylinder fully instroked. Circuit waits for operator to press button again to start the next cycle.

Task 9 — Semi-automatic circuit

9a. Describe the complete operating cycle of a semi-automatic pneumatic circuit. Include reference to: the pilot signal, how main air causes the outstroke, what triggers the instroke, and what happens at the end of the cycle.

9b. Explain the difference between a manual circuit and a semi-automatic circuit. Give one advantage of using a semi-automatic circuit in a production environment.

Section 10: Fully Automatic Circuit

A **fully automatic circuit** repeats its cycle continuously without any manual input after the initial start signal.



No Pressure Sim — Fully Automatic Circuit

This interactive simulator is in the online course notes. Scan the QR code, or use a device, to open it.

This automatic cycle would continue until main air is cut off.

Fully automatic vs semi-automatic: In a semi-automatic circuit, only one limit valve is used so the cylinder stops and waits at the instroked position. In a fully automatic circuit, a second limit valve at the instroked end immediately sends a pilot signal to trigger the next outstroke.

Task 10 — Fully automatic circuit

10a. Explain the role of each limit valve in a fully automatic circuit. What would happen if the limit valve at the retracted position were removed?

10b. A fully automatic circuit is used in a bottling plant to cap bottles on a conveyor. Describe how the circuit could be modified to allow an operator to stop the machine safely at the end of a cycle (not mid-stroke).

Section 11: Time Delay Circuit

A **time delay valve** introduces a controllable delay between an input signal and an output action.



No Pressure Sim — Time Delay Circuit

This interactive simulator is in the online course notes. Scan the QR code, or use a device, to open it.

The circuit consists of a uni-directional restrictor feeding a **reservoir**, which builds up main air pressure slowly until it is high enough to pilot (switch) a 3/2 output valve. By adjusting the restrictor, the fill time of the reservoir — and therefore the length of the delay — can be changed. Increasing the size of the reservoir will also increase the delay time, as more volume must be filled before the pilot threshold is reached. When the input signal is removed, the reservoir vents quickly back through the UDR and the timer resets immediately.

Task 11 — Time delay circuit

11a. Describe how a pneumatic time delay circuit produces a time delay. Include reference to the restrictor, reservoir, and pilot-operated output valve.

11b. A time delay circuit is set to delay for 3 seconds. An engineer wants to increase the delay to 8 seconds. Explain two ways this could be achieved.

11c. Use the simulator or physical components to build a time delay circuit. Adjust the UDR to make the delay longer and shorter. Describe what you observe and explain why adjusting the restrictor changes the delay time.

11d. Describe one practical application of a time delay circuit in an industrial or domestic context. Explain what the delay achieves in your chosen example.

11e.  Draw the two key components that generate a time delay in a pneumatics circuit. Label the components and the direction of air flow. Draw your answer in your jotter.

Section 12: Electrical Control of Pneumatics

Every circuit so far has been switched by hand, by a pilot air signal, or automatically by another part of the pneumatic circuit. A pneumatic valve can also be switched **electrically**, using a **solenoid** actuator — a coil that, when energised, moves the valve's spool. This is how pneumatic systems are controlled by microcontrollers, Programmable Logic Controllers (PLCs) and other electronic systems.

A solenoid-actuated valve is connected to an electrical control system (such as a microcontroller or PLC). When the control system sends an electrical signal to the solenoid, it switches the valve and main air flows to the cylinder. When the electrical signal is removed, the spring returns the valve to its normal state and the cylinder instrokes. This allows a pneumatic circuit to be controlled entirely by software or by electronic sensors such as light gates or pressure switches.

Key point: A solenoid valve is the bridge between the electronic and pneumatic worlds. Everywhere you'd otherwise need a person to press a button or a mechanical pilot signal, a solenoid lets a microcontroller, sensor, or PLC do the switching instead.

Task 12 — Electrical control

12a. Explain the difference between a pilot actuator and a solenoid actuator on a valve. In which type of circuit would each be used?

12b. Give one example of a sensor that could be used to trigger a solenoid valve electrically, and describe the pneumatic action it would cause.

12c.  Draw the full solenoid circuit diagram: a solenoid-actuated 3/2 valve connected to a single acting cylinder, labelling the electrical signal, main air and exhaust. Draw your answer in your jotter.

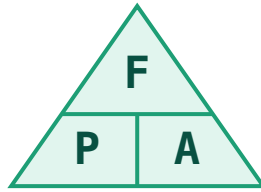
Section 13: Calculating Cylinder Force

The force a pneumatic cylinder can produce depends on two things: the **air pressure** in the circuit, and the **area** of the piston that the pressure is acting on.

$$F = P \times A$$

$$\text{Force (N)} = \text{Pressure (N/mm}^2\text{)} \times \text{Area (mm}^2\text{)}$$

You can rearrange this formula to find any of the three quantities. The **FPA triangle** is a useful memory aid:



Cover the quantity you want to find — the other two show how to calculate it.

$$F = P \times A$$

$$P = F / A$$

$$A = F / P$$

Watch your units: In National 5 Engineering Science, pressure is given in **N/mm²** and diameters in **mm** — this means **no conversion is needed**: work in millimetres throughout and your force comes out directly in Newtons. (You may see pressure quoted elsewhere in bar or pascals, but you won't be asked to convert between these units.)

Piston area and rod area

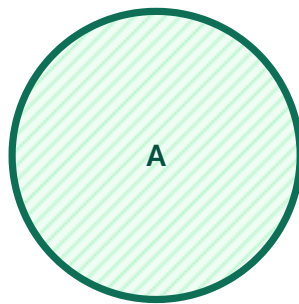
A single acting cylinder only has main air pushing on **one** side of the piston (the outstroke). The full circular face of the piston is available to push against, so the effective area is simply the **piston area**:

$$A = \pi \times \frac{D^2}{4}$$

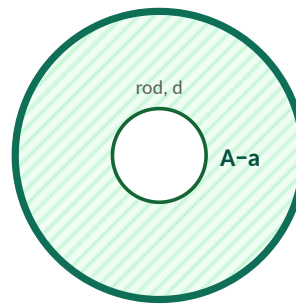
where D is the diameter of the piston

Engineers use the **diameter** of the piston (rather than its radius) mainly because diameter is what's easy to measure in the workshop — you can measure straight across a piston or a rod with a pair of **callipers** (or a vernier calliper) far more easily and accurately than you could measure to the centre for a radius.

A double acting cylinder has main air pushing on **both** sides of the piston, one side at a time. On the outstroke, the full piston area is available — exactly like a SAC. On the instroke, main air acts on the other side of the piston, but the **piston rod** takes up part of that face, so the effective area is the piston area **minus the rod area**.



Outstroke — full piston area
Main air pushes on the whole piston face:
piston area = $\pi \times D^2 / 4$



Instroke — piston area minus rod area
rod area = $\pi \times d^2 / 4$
effective area = piston area – rod area

What each letter means

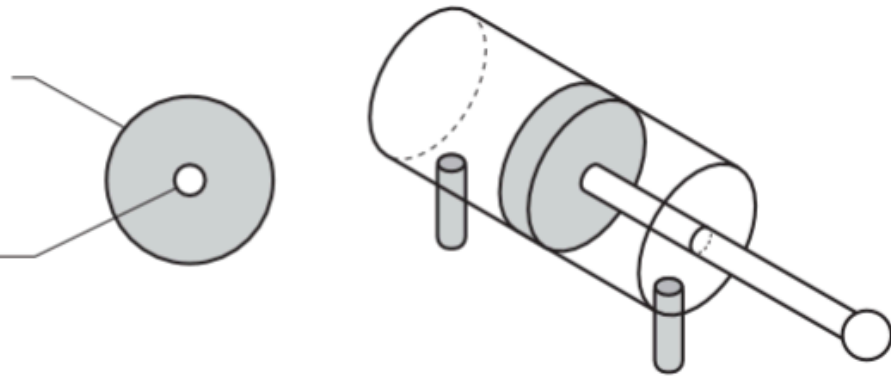
- D** diameter of the **piston**
- d** diameter of the **rod**
- A** **piston area** — the full circular face of the piston
- a** **rod area** — the cross-section of the piston rod

Key point: Because the effective area on the instroke (piston area minus rod area) is smaller than the piston area used on the outstroke, a DAC always produces **less force on the instroke** than on the outstroke, for the same supply pressure.

Worked Example 1 — outstroking and instroking force

Piston
70mm diameter

Piston rod
10mm diameter



The cylinder above has a piston diameter of 70 mm and a piston rod diameter of 10 mm. It is supplied at 0.5 N/mm². Calculate the outstroking force and the instroking force.

OUTSTROKING FORCE

- 1 Piston area: $A = \pi \times 70^2 / 4 = 3848 \text{ mm}^2$.
- 2 Force: $F = P \times A = 0.5 \times 3848 = \mathbf{1924 \text{ N}}$.

INSTROKING FORCE

- 1 Rod area: $a = \pi \times 10^2 / 4 = 78.5 \text{ mm}^2$.
- 2 Effective area: piston area – rod area = $3848 - 78.5 = 3770 \text{ mm}^2$.
- 3 Force: $F = P \times A = 0.5 \times 3770 = \mathbf{1885 \text{ N}}$.

Worked Example 2 — calculating piston diameter

A door requires a force of 354 N to slide it open. A single acting cylinder supplied at a pressure of 0.5 N/mm² controls the operation. Calculate the minimum diameter of piston required.

PISTON DIAMETER

- 1 Rearrange the formula to find area: $A = F / P$.
- 2 Area: $A = 354 / 0.5 = 708 \text{ mm}^2$.
- 3 Rearrange the area formula to find diameter: $D = \sqrt{(4A / \pi)}$.
- 4 Diameter: $D = \sqrt{(4 \times 708 / \pi)} = \sqrt{901.4} = \mathbf{30.0 \text{ mm}}$.

Worked Example 3 — calculating supply pressure

A stamping machine exerts a force of 454 N with a piston diameter of 34 mm. Calculate the air pressure required.

SUPPLY PRESSURE

- 1 Piston area: $A = \pi \times 34^2 / 4 = 907.9 \text{ mm}^2$.
- 2 Rearrange the formula to find pressure: $P = F / A$.
- 3 Pressure: $P = 454 / 907.9 = 0.5 \text{ N/mm}^2$.

Task 13 — Force calculations

13a. A SAC has a piston diameter of 20 mm and is supplied at 0.4 N/mm^2 . Calculate the outstroke force in Newtons. Show your working.

13b. A DAC has a piston diameter of 40 mm and a piston rod diameter of 12 mm, supplied at 0.6 N/mm^2 . Calculate (i) the outstroking force and (ii) the instroking force. Show your working for both, including the piston area and the rod area.

13c. Explain, in terms of piston area and rod area, why a DAC's instroking force is always lower than its outstroking force at the same supply pressure.

13d. A single acting cylinder must produce a force of 250 Newtons when supplied at 0.5 N/mm^2 . Calculate the minimum piston area required. Show your working.

13e. A double acting cylinder has a piston diameter of 32 mm. Calculate the supply pressure required to produce an outstroke force of 643 Newtons. Show your working.

Section 14: Design Challenges

The questions in Sections 1–12 checked that you understand each circuit individually. The three challenges below work differently — each one gives you a real-world context and a written **specification**, the same way an assignment does. Nobody will show you the circuit first: you have to work out which components to use and how to connect them, then build or simulate your design and check it actually meets the spec.

How to approach these: Read the whole specification before you draw anything. Underline the key words that tell you which component you need ("*either... or*" usually means a shuttle valve; "*slowly and smoothly*" means a restrictor on the exhaust; "*after a delay*" means a time delay circuit). Sketch a rough plan first, then tidy it into a proper circuit diagram.

Design Challenge 1 — Car wash spray arm

A car wash has a spray arm, powered by a double acting cylinder, that swings out over each car. If the arm swings out too fast it splashes water everywhere, so it needs to move out slowly and under control. Once the wash is finished, the arm should retract quickly out of the way so the next car can pull in.

- i Pressing push-button A must cause the cylinder to outstroke **slowly and smoothly**.
- ii Pressing push-button B must cause the cylinder to instroke at normal (unrestricted) speed.
- iii The circuit must not restrict the main air supply into the cylinder at any point.

Design Challenge 1

DC1a. Design a pneumatic circuit to meet the specification above. You must identify every component, valve and actuator, and show the connections between them. Draw your design in your jotter.

DC1b. Build or simulate your design and test it. Describe what actually happened when you pressed each button.

DC1c. Evaluate your circuit against each point of the specification. For each point, state whether it is met and justify your answer.

Design Challenge 2 — Greenhouse roof vent

A commercial greenhouse has a long roof vent, opened and closed by a double acting cylinder, that needs to be controlled from **either end** of the greenhouse — whichever member of staff is closest should be able to open it, without needing to know whether someone at the other end is also about to use it.

- i The cylinder must outstroke (open the vent) when push-button valve 1 or push-button valve 2 is actuated, whether the other is actuated or not.
- ii The cylinder must instroke (close the vent) when a third push-button valve is actuated.

Design Challenge 2

DC2a. Design a pneumatic circuit to meet the specification above. You must identify every component, valve and actuator, and show the connections between them. Draw your design in your jotter.

DC2b. Build or simulate your design. Test it with valve 1 alone, valve 2 alone, and both together., noting what you observed for each test.

DC2c. A colleague suggests replacing the shuttle valve with a simple T-piece, since "it also joins two pipes together." Explain why this would not work.

DC2d. Can you suggest a design improvement? This should make the solution better than is specified.

Design Challenge 3 — Car park barrier

An automatic barrier at a car park entrance is raised by a double acting cylinder. When a driver presses a button, the barrier should raise. Once fully raised, it should stay up just long enough for the car to pass, then lower automatically by itself — without the driver having to press anything else.

- i Pressing the entry button must cause the cylinder to outstroke, raising the barrier.
- ii Once the barrier reaches the fully raised position, an **adjustable time delay** of a few seconds must occur.
- iii After the delay, the cylinder must instroke automatically, lowering the barrier, with no further input from the driver.

Hint: think about how you detected "fully raised" in the semi-automatic and fully-automatic circuits in Sections 9 and 10, and how you produced a delay in Section 11 — this challenge combines both ideas.

Design Challenge 3

DC3a. Design a pneumatic circuit to meet the specification above. You must identify every component, valve and actuator, and show the connections between them. Draw your design in your jotter.

DC3b. Build or simulate your design and test it. Describe what happened.

DC3c. Evaluate your circuit against each point of the specification, stating whether it is met and justifying your answer. Then suggest one improvement that would make the barrier safer for a car that is still underneath it when the delay ends.